

ADHD Brain Connectomics: Plain-English Neuroimaging Guide

Why the Default Mode Network Fails to Suppress During Tasks | Framework: ADHD-200 Consortium & Functional MRI Brain-Mapping

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Clinical Resource Packet

CHILD'S NAME	AGE / GRADE	DATE	COMPLETED BY

Understanding the 3 Brain Networks in ADHD		
BRAIN NETWORK	NEUROTYPICAL FUNCTION	WHAT HAPPENS IN THE ADHD BRAIN
Default Mode Network (DMN)	Responsible for mind-wandering, daydreaming, and internal self-reflection. When a person focuses on a math problem, the brain automatically shuts down the DMN.	Fails to Turn Off: In ADHD, the DMN stays active even during difficult tasks, intruding into consciousness with thoughts about lunch, video games, or random ideas.
Central Executive Network (CEN)	Governs working memory, goal-directed planning, task initiation, and error correction.	Under-Activated: Struggles to maintain sustained signal strength unless external dopamine stimulation (high interest, novelty, urgency) is present.
Salience Network (SN)	The "switchboard operator" that decides which stimuli deserve immediate attention (e.g. teacher's voice vs. pencil clicking).	Broad Filter: Treats extraneous classroom background noises (fluorescent hum, car outside) with equal urgency as the teacher's instructions.

Reframing for Parents: Your child is not daydreaming to defy you. Their Default Mode Network is physically failing to suppress due to low dopamine and norepinephrine signaling in the prefrontal cortex.